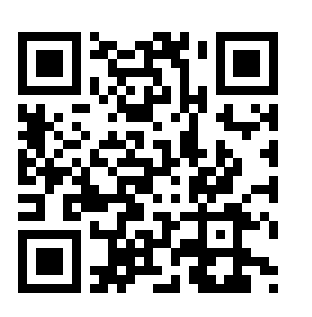


Where Fractals Just Touch

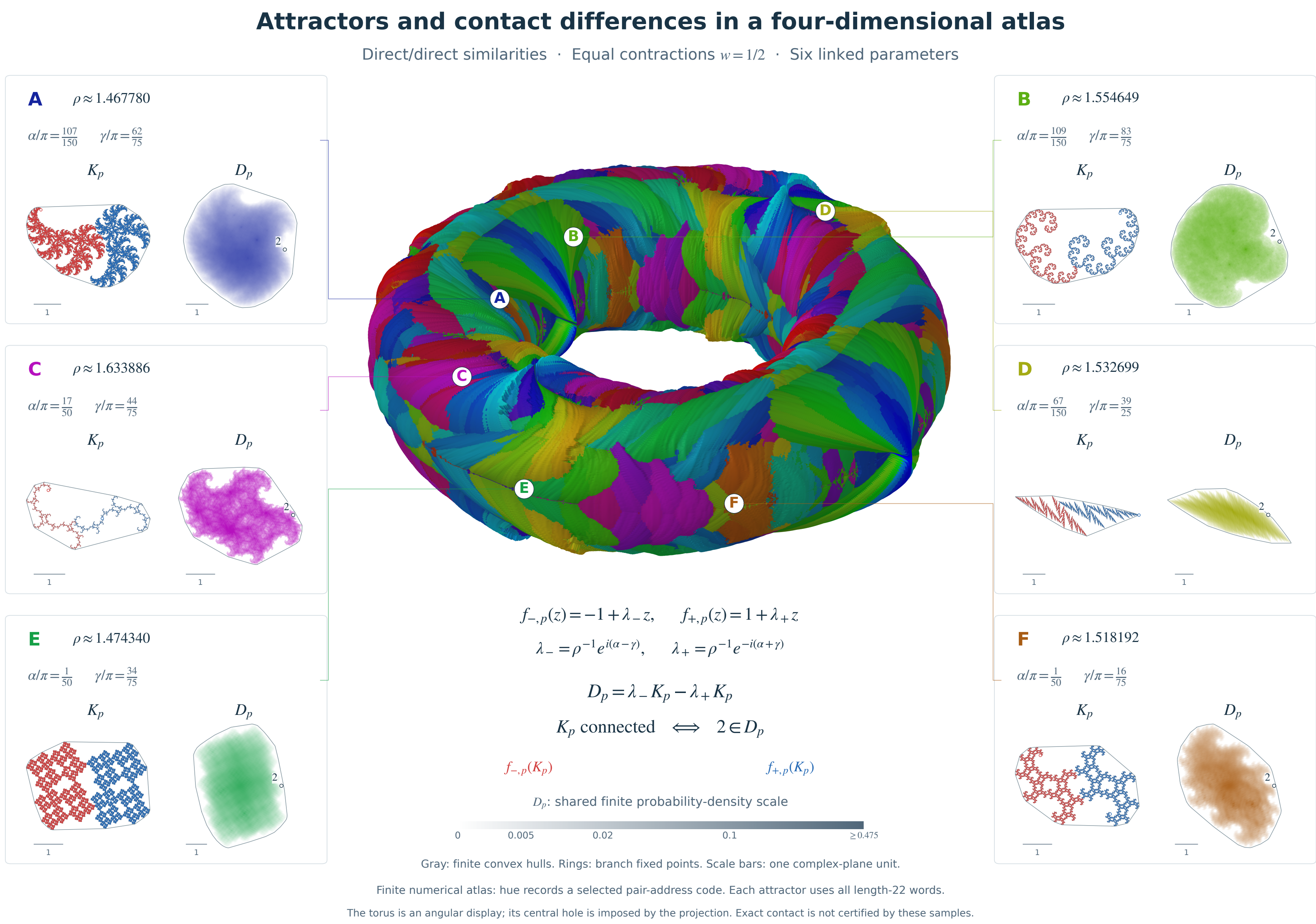
A Four-Dimensional Atlas of Connectedness and Zipper Stability

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From four-dimensional parameters to stable fractal curves.



1. One atlas in exterior parameters

Two complex parameters with $|c_1|, |c_2| > 1$ define

$$S_1(z) = -1 + c_1^{-1} J_{\kappa_1}(z), \quad S_2(z) = 1 + c_2^{-1} J_{\kappa_2}(z),$$

where $J_+(z) = z$ and $J_-(z) = \bar{z}$. Direct/direct, direct/opposite and opposite/opposite maps give three orientation types. For $K = S_1(K) \cup S_2(K)$, connectedness becomes

$$2 \in c_1^{-1} J_{\kappa_1}(K) - c_2^{-1} J_{\kappa_2}(K).$$

The atlas separates phases, magnitude and weight:

$$\rho = \frac{2|c_1||c_2|}{|c_1| + |c_2|}, \quad w = \frac{|c_2|}{|c_1| + |c_2|}.$$

A three-dimensional angular projection shows the surfaces; colour records the fourth coordinate w .

2. Curve families inside the connectedness loci

A binary zipper joins 0 to 1 through a vertex q . Its plane orientation and endpoint reversal (e_1, e_2) select a smooth two-dimensional surface inside the atlas. Its stable set consists of embedded canonical curves:

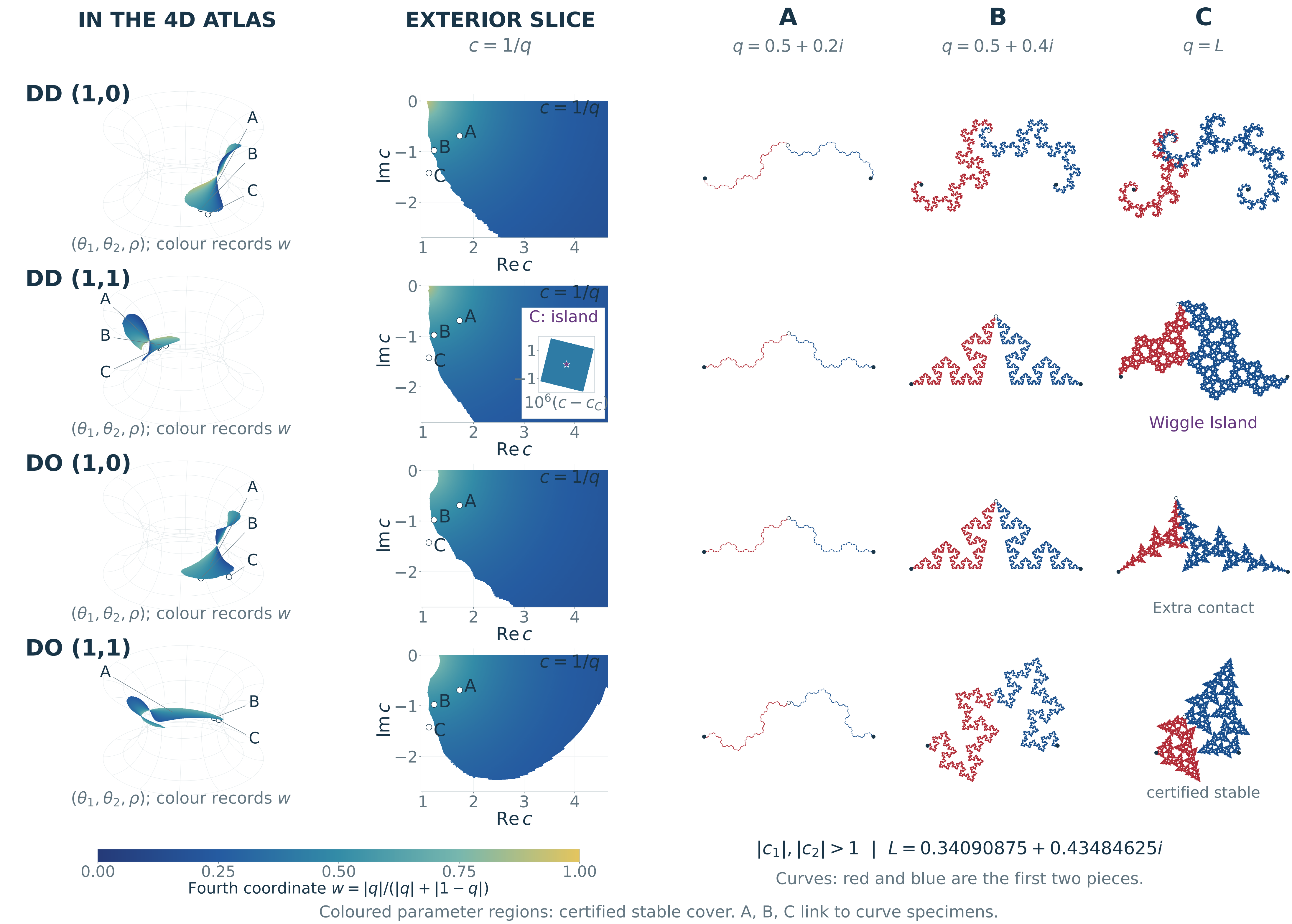
$$q \in \mathcal{S}_{\kappa,e} \iff F_1(K_q) \cap F_2(K_q) = \{q\}.$$

Finite endpoint-return identities certify whole regions of stable parameters. These regions lie in the band

$$2\sqrt{w^2 + (1-w)^2} < \rho \leq 2.$$

Below, four selected families share three parameter probes. The DD(1,1) surface contains the Koch/wiggle island [3]. The exterior slice coordinate is $c = 1/q$; both similarity parameters have modulus greater than one.

3. Four stable surfaces, twelve linked curve examples



Explore the atlas: complextrees.com/4D/

Companion manuscript: B. Espigulé, *Four-Dimensional Connectedness Loci and Structural Stability for Planar Binary Similarities* (2026).

References. [1] B. Espigulé. *Families of connected self-similar sets generated by complex trees*. arXiv:1902.11282 (2019).
[2] B. Espigulé Pons. *Collinear Fractals and Connectedness Loci: Topology, Finite Capture, and Restricted Polynomial Roots*. Doctoral thesis (submitted), Universitat de Girona, 2026. complextrees.com/research/thesis/.
[3] D. Calegari. *Wiggle Island*. arXiv:2205.11442 (2022; revised 2026).

Supported by the Spanish Ministerio de Ciencia, Innovación y Universidades through project PID2023-146424NB-I00.